

BAYEVO, Russian Federation

WMO: 298270

Lat: 53.2781N

Lon: 80.7755E

Elev: 400

StdP: 14.48

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-27.1	-38.5	0.6	-32.6	-32.3	0.9	-26.6	29.7	16.8	25.9	17.1	3.4	250	0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	86.8	65.7	83.7	64.9	80.9	63.7	69.0	80.2	67.4	78.6	66.0	77.0	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	95.6	73.5	63.5	89.1	72.2	61.8	83.8	70.8	33.5	80.5	32.2	78.7	31.0	77.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.6	21.0	18.4	DB	-39.9	91.9	8.2	3.2	-45.8	94.2	-50.5	96.0	-55.1	97.8	-61.0	100.1	
			WB	-39.4	72.6	7.1	3.3	-44.5	74.9	-48.7	76.8	-52.7	78.6	-57.8	81.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.5	-0.2	4.5	19.1	41.0	55.3	65.5	67.9	63.3	52.2	39.0	21.4	7.5
(7)		DBStd	26.40	15.57	13.96	12.48	10.20	9.17	6.85	5.64	6.28	7.75	8.85	14.09	14.83
(8)		HDD50	6745	1558	1275	957	304	55	2	0	1	65	353	857	1318
(9)		HDD65	10684	2023	1695	1422	722	317	77	36	106	390	806	1307	1783
(10)		CDD50	1833	0	0	0	33	220	468	555	415	130	12	0	0
(11)		CDD65	297	0	0	0	0	17	93	127	55	5	0	0	0
(12)		CDH74	3010	0	0	0	13	267	854	1060	683	133	0	0	0
(13)		CDH80	798	0	0	0	1	58	229	285	187	38	0	0	0
(14)	Wind	WSAvg	7.6	7.6	7.5	8.1	8.5	8.4	7.0	6.0	6.3	6.7	7.5	8.6	8.7
(15)	Precipitation	PrecAvg	13.7	0.6	0.5	0.6	0.8	1.2	1.8	2.6	1.4	1.2	1.1	1.1	0.9
(16)		PrecMax	19.4	1.5	1.3	1.2	2.3	3.5	4.3	5.2	3.5	2.8	3.1	2.2	1.6
(17)		PrecMin	7.1	0.1	0.0	0.0	0.0	0.1	0.4	0.4	0.1	0.2	0.3	0.4	0.3
(18)		PrecStd	3.1	0.3	0.3	0.3	0.5	0.8	1.0	1.3	0.8	0.7	0.7	0.4	0.3
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	35.0	46.8	75.6	87.2	89.6	89.9	90.2	84.9	69.1	50.4	36.3
(20)			MCWB	32.9	32.7	40.0	55.8	62.3	65.3	68.3	66.5	60.8	53.1	44.8	33.3
(21)		2%	DB	30.4	31.5	41.3	68.5	80.8	85.8	86.1	84.9	77.5	62.8	46.1	33.2
(22)			MCWB	28.7	29.6	36.5	52.8	58.4	64.4	67.2	66.1	58.5	50.1	40.9	31.3
(23)		5%	DB	26.5	28.7	38.2	62.6	76.8	82.4	83.2	81.1	71.9	58.0	41.3	30.2
(24)			MCWB	24.8	26.9	34.5	50.0	57.5	64.1	66.3	64.0	56.4	47.8	37.6	28.6
(25)		10%	DB	22.6	24.9	35.4	57.4	72.1	79.1	80.1	77.5	67.3	52.7	37.5	27.3
(26)			MCWB	21.2	23.3	32.7	47.2	56.3	63.2	65.6	62.6	54.4	45.3	35.0	25.7
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.4	33.2	40.9	57.4	65.8	70.0	72.7	70.7	63.2	54.5	46.1	33.7
(28)			MDCB	34.8	34.4	45.9	72.4	80.7	81.5	82.3	81.5	76.8	65.5	49.8	35.4
(29)		2%	WB	28.8	29.9	37.0	53.8	61.7	67.3	69.8	67.5	59.9	51.2	41.1	31.4
(30)			MDCB	30.2	31.4	40.5	65.5	75.4	79.2	80.5	80.9	74.1	60.6	45.3	32.9
(31)		5%	WB	25.0	26.9	34.9	50.8	59.4	65.8	68.3	65.6	57.8	48.5	38.0	28.7
(32)			MDCB	26.4	28.6	37.8	61.0	72.5	77.7	78.5	78.3	70.1	56.7	41.5	30.1
(33)		10%	WB	21.2	23.4	32.8	47.8	57.3	64.4	66.9	63.8	55.4	45.8	34.9	25.7
(34)			MDCB	22.6	24.8	35.2	57.0	69.8	75.8	76.8	75.4	65.6	52.3	37.4	27.3
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.3	18.0	17.7	18.3	21.0	19.2	18.6	21.3	20.6	16.1	11.9	14.8
(36)			MCDBR	16.7	15.3	16.7	26.9	28.3	25.2	23.3	28.0	30.7	26.8	14.0	15.7
(37)			MCWBR	16.0	14.3	12.9	13.8	11.3	9.0	8.5	11.3	14.0	15.4	10.5	15.1
(38)		5% WB	MCDBR	17.0	15.2	14.8	25.3	23.0	20.6	19.2	24.0	26.5	25.0	13.3	15.0
(39)			MCWBR	16.4	14.4	11.7	14.1	10.9	9.0	8.8	10.7	13.0	15.1	10.3	14.6
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.333	0.384	0.408	0.410	0.410	0.434	0.416	0.373	0.355	0.336	0.304	
(41)		taud	2.020	1.958	1.910	2.137	2.194	2.279	2.210	2.234	2.302	2.238	2.076	1.974	
(42)		Ebn at Noon	199	230	244	259	266	267	257	255	252	226	182	166	
(43)		Edh at Noon	24	36	47	43	43	40	42	39	32	27	23	21	
(44)	All-Sky Solar Radiation	RadAvg	357	698	1164	1499	1829	1979	1788	1547	1099	580	286	242	
(45)		RadStd	30	44	100	99	156	147	118	111	107	79	44	35	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	-1.12	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	-0.86	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page